

PRMs Estimate Step Correctness; Rollback Requires Counterfactual Necessity

Anonymous ACL submission

Abstract

PRMs evaluate whether reasoning steps are correct, whereas rollback requires predicting whether deleting a step will improve or harm the downstream outcome. We find that correctness signals provide useful information about removal effects but do not determine the outcome of intervention. Across 600 PRM800K steps balanced across correctness ratings, deleting rating=-1 steps improves accuracy by 22.0 percentage points on average, while rating=0 and rating=+1 groups show near-zero average effects. However, removal effects remain heterogeneous within each correctness group, showing that correctness alone cannot serve as a rollback criterion. Correctness signals can nominate suspicious steps, but safe rollback requires intervention-aware estimation of removal effects.

1 Introduction

Process reward models (PRMs) provide step-level supervision for large language model (LLM) reasoning by estimating whether an intermediate step is correct. Their scores are also natural inputs to reasoning interventions such as selection, revision, and rollback. Yet evaluation and intervention optimize different objectives. A correctness evaluator answers, “Is this step valid?” A rollback mechanism asks, “Would the model perform better without this step?” These questions are related, but not equivalent. Existing process supervision methods are designed for ranking reasoning quality, whereas rollback requires estimating the marginal effect of intervention; PRM scores solve the former objective but are not designed to directly measure the latter.

Rollback is therefore counterfactual: it asks what would happen if this step disappeared, a question not answered by observing the original trajectory. We call the corresponding property *counterfactual necessity*: whether keeping a step is necessary

for preserving the downstream outcome. Operationally, we measure its complement, counterfactual removability, by deleting the step, regenerating the continuation, and comparing the resulting answer accuracy. A locally correct step can be removable, while an incorrect step can be useful or can introduce a constraint from which the generator fails to recover. Thus rollback safety depends on the step-generator interaction after intervention, not on the original step judgment alone.

We first establish the correctness-necessity mismatch. We then analyze possible explanations for beneficial deletions, including whether locally plausible steps introduce harmful constraints. Finally, we study the implication for rollback selection under a predefined risk budget.

We answer these questions with a counterfactual deletion benchmark over PRM800K trajectories. For each target step, we compare the original continuation with a target deletion and, when available, a position- and length-matched placebo deletion. This design treats deletion as a measurement tool for counterfactual necessity, rather than as a pruning objective. We then evaluate correctness-based and intervention-aware rollback policies as a diagnostic policy evaluation, not as a new rollback algorithm.

Our central finding is that correctness and removability are correlated but non-equivalent (Cramér’s $V = 0.53$; 34.5% of steps are off the expected diagonal), with the primary evidence in Table ?? . We then examine why some removals help through harmful-constraint analysis and placebo controls (Figure ?? and Table ??), and evaluate the practical implication for rollback policy selection (Figure ??). This framing preserves the value of PRMs as correctness evaluators while identifying the additional property required for intervention.

Figure ?? makes the distinction concrete without treating either outcome as a PRM error: the PRM supplies a local correctness judgment

ment, while the intervention reveals whether the step is necessary for the downstream answer.

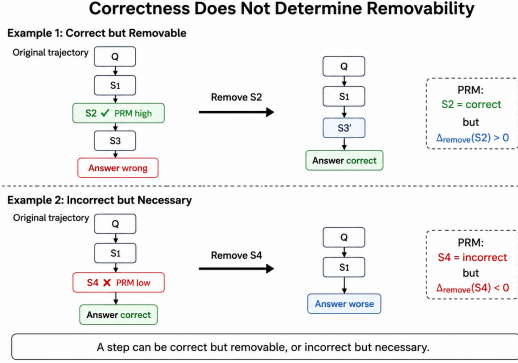


Figure 1: Correctness Is Not Necessity. A PRM can judge a step correct even when removing it improves the answer, and judge another step incorrect even when removing it hurts. Δ_{remove} denotes the change in final-answer accuracy after deleting the step and regenerating the continuation.

2 Framework: From Step Correctness to Counterfactual Necessity

PRM scores estimate the local correctness of individual reasoning steps in an existing trajectory. Rollback requires a different property: whether the step is necessary for the downstream outcome. To formalize this distinction, we separate local correctness, semantic contribution, and counterfactual necessity. We measure necessity through its intervention counterpart, counterfactual removability: a step is more removable when deleting it preserves or improves the outcome, and more necessary when deletion harms it. These properties answer different questions about the same step rather than competing definitions of quality.

2.1 Correctness, Contribution, and Counterfactual Necessity

A reasoning step can be evaluated from multiple perspectives that are often conflated in process supervision. We distinguish three properties.

Local correctness. Local correctness measures whether a reasoning step is valid according to process supervision. In PRM-based evaluation, each intermediate step receives a score or label indicating whether the step is incorrect, uncertain, or correct. This evaluation considers the step within the original reasoning trajectory and does not measure the effect of modifying the step.

Semantic contribution. Semantic contribution describes whether a reasoning step provides useful information for subsequent reasoning. A step

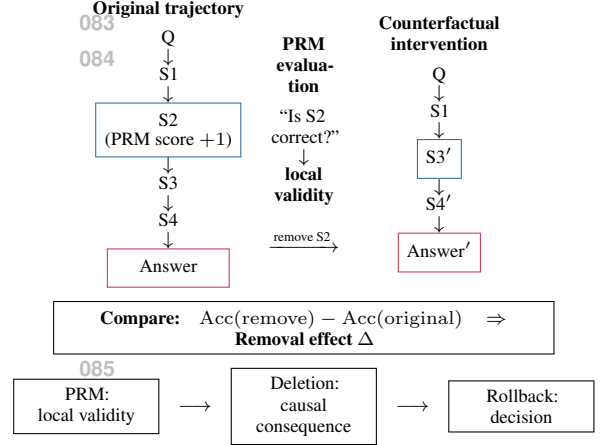


Figure 2: Counterfactual Deletion Measures Necessity Beyond Step Correctness. PRM evaluation asks whether the observed step is locally valid; deletion removes that step, regenerates the continuation, and measures whether retaining it is necessary for the downstream outcome. Deletion is a measurement intervention, not a pruning objective.

may contribute positively by introducing necessary information, have limited impact because it is redundant, or negatively influence future reasoning by introducing misleading constraints. Therefore, semantic contribution captures the role of a step in the reasoning process rather than its correctness alone.

Counterfactual necessity. Counterfactual necessity asks whether the final outcome depends on retaining a reasoning step. We operationalize it through counterfactual removability: the target step is removed, the remaining trajectory is regenerated, and the resulting outcome is compared with the original trajectory. A step is necessary when removal damages the outcome, and removable when removal preserves or improves it.

A correct step may still be removable if it does not affect the final solution, while an incorrect step may be necessary if later reasoning relies on it. Therefore, correctness and necessity represent different properties of reasoning steps, and deletion is the measurement intervention used to distinguish them.

Figure ?? illustrates the relationship between local correctness and counterfactual necessity.

2.2 Measuring Counterfactual Necessity

Given a reasoning trajectory consisting of intermediate steps followed by a final answer, we measure counterfactual necessity through controlled removal. For a target step s_i , we remove the step and regenerate the subsequent reasoning process

from the same position. The resulting trajectory is then evaluated using the final-answer outcome.

We define the removal effect of step s_i as:

$$\Delta_i = \text{Acc}(\text{Delete}(s_i)) - \text{Acc}(\text{Control}). \quad (1)$$

where $\text{Acc}(\text{Control})$ denotes the accuracy of the original trajectory and $\text{Acc}(\text{Delete}(s_i))$ denotes the accuracy after removing step s_i and regenerating the continuation.

A positive Δ_i indicates that removing the step improves the final outcome, suggesting that the step is removable and may introduce a harmful constraint. A negative Δ_i indicates that removing the step damages performance, suggesting that retaining the step is necessary for this generator and trajectory.

However, deletion introduces a confounding factor: removing any step changes the generation state and may produce improvements unrelated to the semantic content of the removed step. To separate step-specific effects from generic regeneration effects, we introduce matched placebo deletions.

2.3 Controlled Deletion Protocol

For each reasoning trajectory, we compare three intervention settings.

Original trajectory. The model follows the complete reasoning trajectory without modification.

Target deletion. A selected reasoning step is removed, and the model regenerates the remaining trajectory from the deletion point. This measures the observed effect of removing the target step.

Placebo deletion. A matched non-target step is removed using the same procedure. This provides an estimate of the performance change caused by regeneration itself rather than by the semantic content of the removed step.

The difference between target deletion and placebo deletion measures the step-specific contribution of the removed reasoning content:

$$\Delta_{\text{sem}} = \Delta_{\text{target}} - \Delta_{\text{placebo}}. \quad (2)$$

This controlled comparison allows us to distinguish step-specific intervention effects from improvements caused only by restarting the generation process. Together, the framework measures counterfactual necessity as an intervention-based property that cannot be inferred solely from local correctness scores.

3 Experimental Setup

We study 600 PRM800K reasoning steps, balanced across PRM ratings $-1/0/+1$, using Qwen3-8B continuations. For each target step s_i , four control and four deletion runs yield the counterfactual effect

$$\Delta_i = \text{Acc}(\text{Delete}(s_i)) - \text{Acc}(\text{Control}).$$

Positive values indicate that removal improves the final answer; negative values indicate that retaining the step is necessary for the evaluated generator and trajectory.

To test whether a gain is specific to the removed step, we additionally use position- and length-matched placebo deletions for 511 steps (1,514 runs). The placebo uses the same truncation and regeneration procedure at a non-target step. All main results use the frozen AI-assisted labels; semantic labels are interpreted as descriptive annotations, not as a substitute for the counterfactual necessity measurement.

Evaluation sequence. We first test the correctness–necessity mismatch, then examine harmful-constraint behavior and placebo controls, and finally evaluate correctness-based rollback selection as an application of the mismatch.

4 Results

The main result is a mismatch between two axes: correctness signals identify suspicious steps, but they do not directly measure the consequence of intervening on those steps.

4.1 Correctness and Counterfactual Necessity Are Different Axes

Table ?? provides the primary test. Low-rated steps are enriched for beneficial removals, but the effect is not determined by correctness: the rating- -1 group gains $+22.00$ pp, whereas rating- 0 and rating- $+1$ are near zero, and the semantic groups span both positive and negative effects. The rating–contribution association is substantial but incomplete (Cramér’s $V = 0.53$; 34.5% off the expected diagonal). Correctness is therefore useful for identifying suspicious candidates, but it does not directly measure rollback necessity.

4.2 Explaining Beneficial Deletions

The mechanism is deliberately stated as an interpretation rather than a universal semantic theory:

Group	Steps	Δ^{tgt} (pp)	95% CI
Overall	600	+7.21	[+3.88, +10.58]
rating = -1	200	+22.00	[+15.41, +28.42]
rating = 0	200	+0.25	[-5.41, +5.90]
rating = +1	200	-0.63	[-5.54, +4.21]
Essential	165	-1.21	[-7.24, +4.78]
Redundant	201	+1.12	[-3.87, +6.06]
Harmful	234	+18.38	[+12.28, +24.44]
-1×Harmful	178	+23.31	[+16.57, +30.24]

Table 1: Primary target-deletion effects. The values show that PRM correctness enriches for suspicious steps but does not determine the intervention effect.

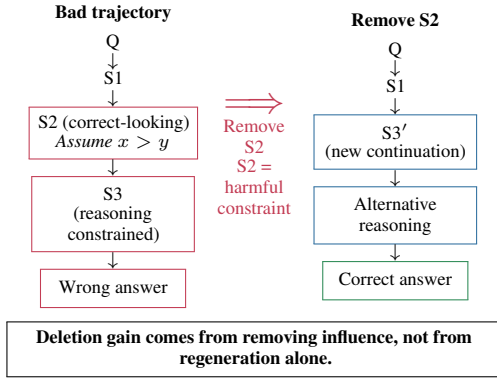


Figure 3: Understanding Beneficial Removals. A locally plausible step can impose a harmful constraint on the continuation. Removing that step can change the downstream path; matched placebos test whether the gain is target-specific.

many beneficial removals are consistent with harmful constraints introduced by earlier reasoning steps that steer the continuation. Figure ?? illustrates this behavior, while the placebo comparison tests whether the improvement is merely arbitrary regeneration.

Group	Δ^{tgt}	$\Delta^{\text{plc}}_{\text{vs C}}$	$\Delta^{\text{plc}}_{\text{own}}$	$\Delta^{\text{sem}}_{\text{DiD}}$ (CI)
Overall (511)	+7.97	-0.57	+0.46	+7.52 [+2.8, +12.4]
rating = -1 (169)	+22.78	+9.47	-0.69	+23.47 [+14.4, +32.5]
rating = 0 (176)	+0.99	-5.63	+2.84	-1.85 [-9.7, +6.0]
rating = +1 (166)	+0.30	-5.42	-0.90	+1.20 [-5.9, +8.4]
Harmful (201)	+18.41	+3.57	-1.74	+20.15 [+12.0, +28.4]
-1×Harmful (151)	+23.84	+9.99	-1.60	+25.44 [+16.1, +34.9]

Table 2: Placebo decomposition. Own-control placebos are near zero, while the target-specific difference-in-differences effect remains positive.

4.3 Rollback Policy Evaluation

Correctness-based ranking is a reasonable way to find suspicious candidates, but it does not estimate whether deleting a candidate will help or hurt. Figure ?? evaluates this application-level consequence: practical rollback selection can benefit from intervention-aware evidence beyond a correctness score alone.

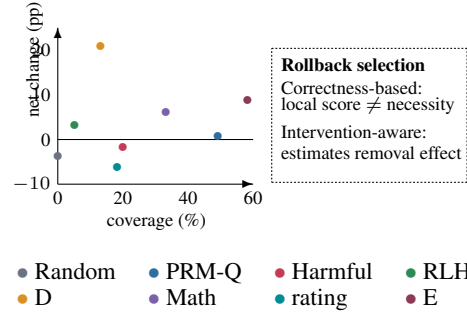


Figure 4: Rollback Policy Evaluation Can Benefit from Intervention-Aware Signals. The plotted positions are schematic and do not add empirical estimates; the figure emphasizes the distinction between a local correctness ranking and an estimate of removal consequences.

Additional analysis. As a scope check, the qualitative correctness–necessity dissociation is reproduced across alternative generators and external trajectories, while additional thinking computation can reduce the observed harmful–constraint pattern. These extensions support the main mismatch without constituting a separate contribution or making the mechanism universal; generator, PRM, annotation, and full policy details are deferred to the appendix.

5 Conclusion

Reasoning interventions require predicting consequences, not only evaluating states. Our study reveals that PRM correctness and rollback necessity are related but distinct: a correctness evaluator asks whether a step is valid, while rollback asks whether the model would do better without it.

Counterfactual deletion provides a direct measurement of necessity and shows that correctness does not determine intervention outcomes; many beneficial deletions are explained by harmful constraints, but this mechanism is not claimed to be universal. PRMs remain valuable for identifying suspicious steps and generating candidates, while practical rollback can benefit from intervention-aware signals beyond local correctness estimation. More broadly, reasoning control should evaluate not only the states models produce, but also the counterfactual consequences of modifying them.

6 Limitations

Our study has three limitations. First, experiments focus on mathematical reasoning trajectories, and generality to other domains remains to be evaluated. Second, counterfactual deletion requires additional

generation and evaluation cost, limiting direct on-line deployment. Third, our analysis studies offline intervention outcomes; practical rollback systems require validation under changing model behaviors and environments.

Future directions. Future work can explore cheaper approximations of counterfactual necessity and intervention-aware rollback in broader reasoning settings.

Ethical Considerations

The study uses the public PRM800K dataset (MIT-licensed problems from MATH) and open-weight models; no personal data is processed. Beyond the authors, four annotators and one adjudicator—unpaid volunteers from the authors’ research group—labeled reasoning steps and paraphrase fidelity for public math problems (about one hour each); they were informed of the study’s purpose and consented, and the annotation sheets collected no personal information. Counterfactual rollback of reasoning steps is an accuracy/safety intervention; our safety framing (harm budgets) is about answer correctness, not user harm. We release artifacts to support reproduction and auditing.

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A Robustness to Continuation Models

The main claim is not tied to one continuation setting. Table ?? reports frozen extension cohorts without pooling them into the primary estimate. The correctness–removability mismatch persists for the no-thinking generators and the external first-error cohort, while additional thinking computation reduces the harmful-constraint pattern. These results are scope checks rather than additional contributions.

Generator / bench- mark		Overall	Target / anchor deletion	Own placebo	DiD semantic
Qwen3-8B, no thinking		+7.2	+23.3	+0.5	+25.4
phi-4, no thinking	+12.4		+33.7	−1.3	+34.9
DeepSeek-R1-Distill, long-CoT	+3.4		+7.0	+8.0	−1.0
Qwen3-8B, thinking	—		−3.0	—	−3.5
ProcessBench, first-error	+24.8		+24.8	−8.4	+35.4

Table 3: Replication summary across continuation models, reasoning modes, and ProcessBench. Values are percentage-point effects from the corresponding frozen extension cohorts; dashes indicate contrasts not reported for that cohort.

B Additional PRM Analyses

Table ?? compares several step-level signals on the identical 600-step cohort. The benefit columns show that some signals provide partial ranking information, but the table does not change the main conclusion that local correctness is not a direct measure of removal consequence.

Signal	Danger AUROC	Danger AUPRC	Benefit AUROC	Benefit AUPRC
Base rate	—	0.15	—	0.24
PRM800K rating	0.56	0.17	0.61	0.34
Qwen2.5-Math-PRM-7B	0.57	0.18	0.61	0.37
Math-Shepherd PRM (7B)	0.47	0.14	0.54	0.26
RLHFlow PRM (8B)	0.51	0.15	0.59	0.28
LLM-judge PRM (Gemini)	0.53	0.16	0.51	0.25
LLM-judge PRM (phi-4)	0.49	0.14	0.48	0.23
PRM disagreement (5 models)	0.53	0.16	0.62	0.34
Step entropy	0.46	0.15	0.48	0.24
Step NLL	0.50	0.15	0.49	0.22
Masked answer-probability drop	0.56	0.19	0.57	0.30

Table 4: Step-level signals versus deletion outcomes. Danger denotes deletion draws that become wrong after a correct control; benefit denotes control-wrong/deletion-correct pairs. AUPRC is reported alongside AUROC because the two outcomes are imbalanced.

C Annotation and Evaluation Details

The 600-step primary cohort uses frozen AI-assisted semantic labels (Essential 165 / Redundant 201 / Harmful 234); these labels are descriptive and do not replace the counterfactual measurement. A blinded 120-step annotation pilot yielded 74

exact-consensus items, with 46 disagreements resolved by an independent adjudicator and pooled agreement $\kappa = 0.40$. The human-only sensitivity check reproduces the harmful-group direction but is too small to establish a full-cohort human ground truth. Separately, a 200-output judge audit reported 93.0% agreement and at most 0.08 pp headline sensitivity under substitution. These audits bound measurement uncertainty without turning the mechanism analysis into a universal semantic claim.

D Full Rollback Policy Values

Table ?? gives the numerical values behind the schematic main-text policy figure. The entries use natural-prevalence weighting and in-sample coverage selection; they are diagnostic comparisons, not deployable safety certificates. Predictor E reads the control-run outcome and is therefore an outcome-informed upper bound.

Policy	Traffic	Budget 1 pp Net / deletion (LCB)	Traffic	Budget 3 pp Net / deletion (LCB)
Random	0	—	0	—
Rating-first	2.4%	+9.8 (+2.0)	12.2%	+1.4 (−1.9)
Harmful-first	3.8%	+17.0 (+1.2)	3.8%	+17.0 (+1.2)
PRM (Qwen2.5-Math)	0.02%	+51.3 (+35.5)	16.3%	+8.2 (−1.8)
PRM (Math-Shepherd)	21.8%	+9.6 (+3.0)	31.3%	+5.2 (−2.8)
PRM (RLHFlow)	40.4%	+5.0 (−0.6)	47.7%	+3.1 (−2.4)
Predictor (D)	66.1%	+4.2 (−0.8)	88.9%	+1.1 (−2.9)
Predictor (E)	68.5%	+4.5 (−0.1)	79.7%	+1.5 (−2.7)
Oracle	96.4%	+3.2 (−0.2)	96.4%	+3.2 (−0.2)

Table 5: In-sample causal certification under natural prevalence. Traffic is the largest coverage whose one-sided 95% cluster lower confidence bound (LCB) clears the stated budget; realized net changes are shown with the LCB in parentheses.